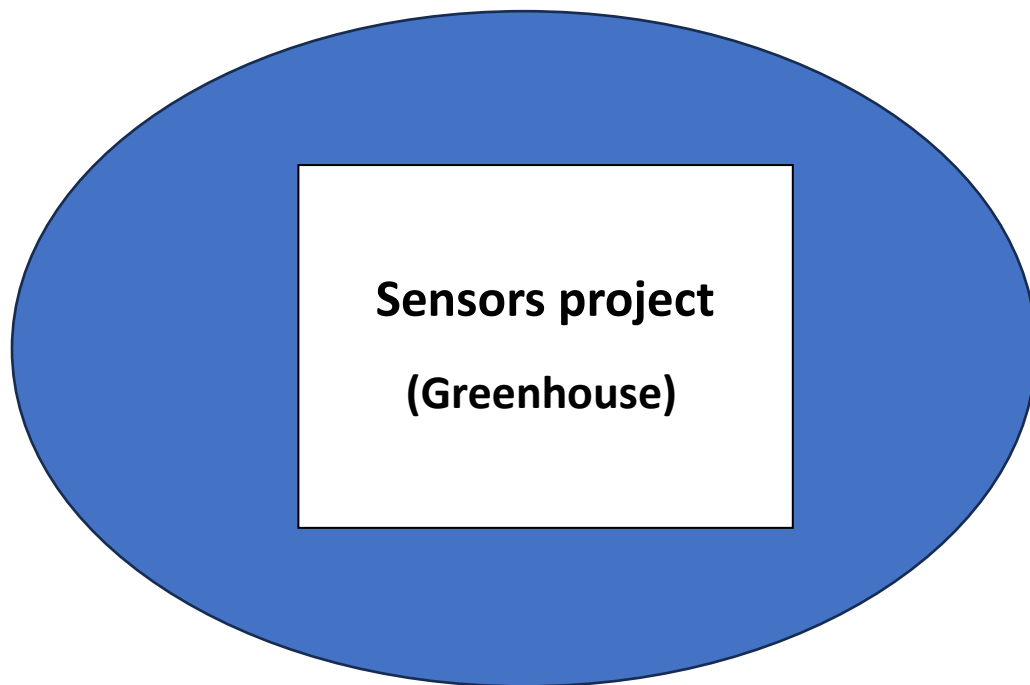


Lebanese University – Faculty of Engineering

Subject: Sensors

Professor: Dr. Mohammad Aoude



Presented by:

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1. Summary of Contribution

I implemented the RFID access-control subsystem using the MFRC522 reader: reading card UIDs, validating them against an authorized list, and driving the green/red status LEDs and LCD messages in `checkRFID()` to show whether the system was armed.

I designed and implemented the six-page LCD interface (temperature/humidity/fan status, flow rate/volume, water tank level, soil moisture/pump status, ultrasonic distance/detection, and sound level), including the button-debounce logic in `handleButton()` and the 500ms refresh cycle in `updateLCD()`

I also took the lead on compiling and formatting the final technical report — writing the introduction, problem statement, and conclusion sections, assembling the architecture diagrams and tables, and proofreading the document end to end before submission.

2. Git Activity

- Add MFRC522 SPI setup and UID reading
- Implement authorized-UID check and LED/LCD feedback
- Build 6-page LCD navigation with push-button cycling
- Add 50ms debounce logic to `handleButton()`
- Add 500ms non-blocking LCD refresh in `updateLCD()`
- Draft introduction and problem statement sections
- Compile final report: architecture diagrams, tables, formatting

3. Reflection

Getting the RFID gating logic right took a few iterations — early on, unauthorized cards were still able to trigger a partial system response before I added a proper state check at the top of the loop. Designing the six LCD pages also pushed me to think about the project from a user's point of view rather than just a sensor's point of view. Pulling the final report together at the end gave me a full picture of how everyone's individual pieces fit into one coherent system.